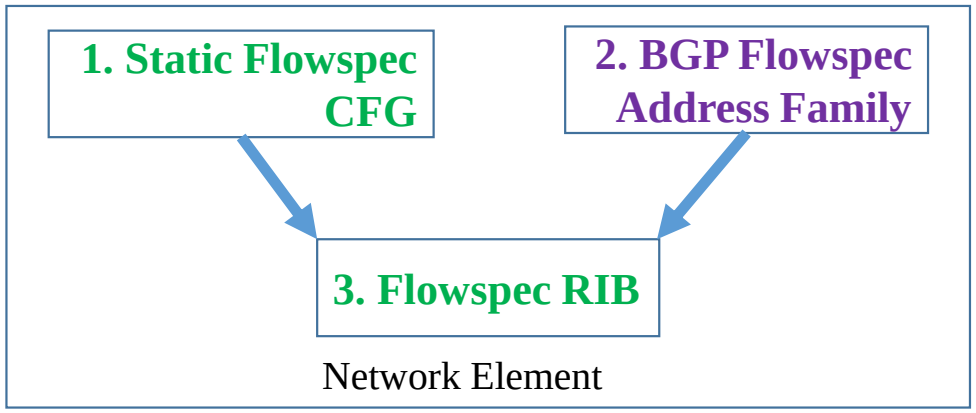
Flowspec and SR Policy YANG

draft-lin-idr-bgp-flowspec-yang
draft-lin-idr-bgp-sr-policy-yang

10-11-2026

Flowspec YANG Components

Flowspec Structure：



1. **Static Flowspec CFG:** is used to configure static flow specification routes which contains filtering rules and corresponding actions, and send static flow specification routes to **Flowspec RIB**.
2. **BGP Flowspec Address Family:** is used to receive flow specification routes via BGP, and send flow specification routes to **Flowspec RIB**.
3. **Flowspec RIB:** is used to store flow specification routes from different protocols, such as Static protocol and BGP.

- **draft-wu-idr-flowspec-yang-cfg**
  - ✓ Defines a YANG data model for Flow Specification implementations.
  - ✓ This YANG model focuses on:
    - **1. Static Flowspec CFG**
    - **3. Flowspec RIB**
  - ✓ It does not include any BGP-related extensions.
- **draft-lin-idr-bgp-flowspec-yang**
  - ✓ Defines a YANG data model for the BGP Flowspec Address Families (IPv4 and IPv6).
  - ✓ This YANG model focuses on:
    - **2. BGP Flowspec Address Family**

BGP Flowspec YANG Model (FSv1)

draft-lin-idr-bgp-flowspec-yang : defines one YANG module "**ietf-bgp-flowspec**".

- ✓ This YANG model extends the BGP Yang modules (ietf-idr-bgp-model) with the BGP FSv1 address families 'ipv4-flowspec' (AFI, SAFI =1, 133) and 'ipv6-flowspec' (AFI, SAFI = 2, 133).
- ✓ It lives under the BGP protocol instance, reusing the BGP YANG model's AFI/SAFI configuration structure, RIB structure, and route annotations.

```
module: ietf-bgp-flowspec

    augment /rt:routing/rt:control-plane-protocols
              /rt:control-plane-protocol
              /bgp:bgp/bgp:global/bgp:afi-safis/bgp:afi-safi:
      +--rw flowspec
        +--rw prefix-limit
        |  ...
        +--rw validate-route-origin?   boolean
```

BGP Flowspec AFI/SAFI Configuration Structure

'**prefix-limit**': is imported from the BGP YANG model's 'mp-all-afi-safi-common' grouping, which limits the number of flowspec routes accepted from a BGP peer.

'**validate-route-origin**': enables the flowspec route validation procedure of RFC 8955, or the revised validation procedure of RFC 9117 when the implementation supports it.

BGP Flowspec YANG Model

BGP Flowspec AFI/SAFI RIB Structure

```
module: ietf-bgp-flowspec
    augment /rt:routing/rt:control-plane-protocols
                /rt:control-plane-protocol
                /bgp:bgp/bgp:rib/bgp:afi-safis/bgp:afi-safi:
       +--ro flowspec
          +--ro loc-rib
           |  +--ro routes
           |     +--ro route* [flow-spec-components]
           |        +--ro flow-spec-components     binary
           |        +--ro active?                  boolean
           |        …
```

```
+--ro neighbors
   +--ro neighbor* [neighbor-address]
      +--ro neighbor-address          inet:ip-address
      +--ro adj-rib-in-pre
      |   +--ro routes
      |       +--ro route* [flow-spec-components path-id]
      |           ...
      +--ro adj-rib-in-post
      |   +--ro routes
      |       +--ro route* [flow-spec-components path-id]
      |           ...
      +--ro adj-rib-out-pre
      |   +--ro routes
      |       +--ro route* [flow-spec-components path-id]
      |           ...
      +--ro adj-rib-out-post
          +--ro routes
              +--ro route* [flow-spec-components path-id]
                  ...
```

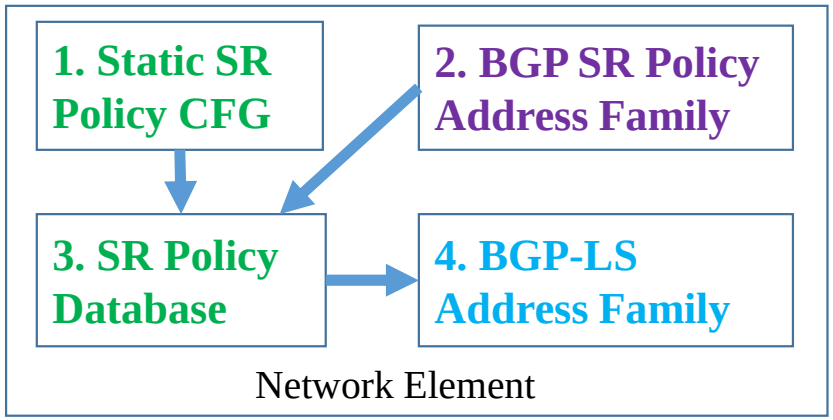
'**loc-rib**' **:** contains the set of BGP flowspec routes selected for local enforcement. The route is keyed by the encoded BGP Flow Specification NLRI ('flow-spec-components');

'neighbors': lists the BGP flowspec routes from neighbors. The route is keyed by 'flow-spec-components' and 'path-id';

'flow-spec-components': will be decoded into match components and traffic filtering actions in the future.

SR Policy YANG Components

SR Policy Structure：



1. Static SR Policy CFG**:** is used to configure static SR Policies which contains Candidate Paths, and send static SR Policies to **SR Policy Database**.
2. **BGP SR Policy Address Family:** is used to receive SR Policy Candidate Paths via BGP, and send SR Policy Candidate Paths to **SR Policy Database**.
3. **SR Policy Database:** is used to store SR Policies from different protocols, such as Static protocol, PCEP and BGP.
4. **BGP-LS Address Family:** is used to advertise SR Policy Candidate Paths via BGP.

- **draft-ietf-spring-sr-policy-yang**
  - ✓ Defines a YANG data model for SR policy framework.
  - ✓ This YANG model focuses on:
    - **1. Static SR Policy CFG** **3. SR Policy Database**
- **draft-aravind-idr-bgp-ls-sr-policy-yang**
  - ✓ Defines a YANG data model for SR Policy Candidate Paths that are advertised using BGP-LS.
  - ✓ This YANG model focuses on:
    - **4. BGP-LS Address Family for SR Policy only**
- **draft-lin-idr-bgp-sr-policy-yang**
  - ✓ Defines a YANG data model for the BGP SR Policy address families.
  - ✓ This YANG model focuses on:
    - **2. BGP SR Policy Address Family**

BGP SR Policy YANG Model

draft-lin-idr-bgp-sr-policy-yang : defines one YANG module "**ietf-bgp-sr-policy**".

- ✓ This YANG model extends the BGP YANG model (ietf-idr-bgp-model) with the BGP SR Policy address families, and defines two BGP SR Policy address family identities: 'ipv4-sr-policy' (AFI, SAFI =1, 73) and 'ipv6-sr-policy' (AFI, SAFI = 2, 73).
- ✓ It lives under the BGP protocol instance, reusing the BGP YANG model's AFI/SAFI configuration structure, RIB structure, and route annotations.

```
module: ietf-bgp-sr-policy

    augment /rt:routing/rt:control-plane-protocols
              /rt:control-plane-protocol
              /bgp:bgp/bgp:global/bgp:afi-safis/bgp:afi-safi:
      +--rw sr-policy
        +--rw prefix-limit
        |  ...
        +--rw propagate-to-ebgp?                boolean
        +--rw ignore-unsupported-subtlvs?   boolean
```

BGP SR Policy AFI/SAFI Configuration Structure

'**prefix-limit**': is imported from the BGP YANG model's 'mp-all-afi-safi-common' grouping, which limits the number of SR Policy routes accepted from a BGP peer.

'**propagate-to-ebgp**': controls the propagation of received SR Policy NLRIs to EBGP neighbors.

'**ignore-unsupported-subtlvs**': provide an option for ignoring unsupported sub-TLVs.

BGP SR Policy YANG Model

BGP SR Policy AFI/SAFI RIB Structure

```
module: ietf-bgp-sr-policy
augment /rt:routing/rt:control-plane-protocols
          /rt:control-plane-protocol
          /bgp:bgp/bgp:rib/bgp:afi-safis/bgp:afi-safi:
    +--ro sr-policy
       +--ro loc-rib
       |  +--ro routes
       |     +--ro route* [distinguisher color endpoint]
       |        +--ro distinguisher          uint32
       |        +--ro color                  uint32
       |        +--ro endpoint               inet:ip-address
       |        +--ro candidate-path
       |           ...
```

'**loc-rib**' **:** contains the set of BGP SR Policy routes selected by the BGP best-path selection algorithm. The route is keyed by the <Distinguisher, Color, Endpoint> tuple of the SR Policy NLRI;

'neighbors': lists the BGP SR Policy routes from neighbors. The route is keyed by 'distinguisher', 'color', 'endpoint', and 'path-id';

```
+--ro neighbors
   +--ro neighbor* [neighbor-address]
      +--ro neighbor-address            inet:ip-address
      +--ro adj-rib-in-pre
      |  +--ro routes
      |     +--ro route* [distinguisher color endpoint path-id]
      |        +--ro distinguisher          uint32
      |        +--ro color                  uint32
      |        +--ro endpoint               inet:ip-address
      |        +--ro path-id                uint32
      |        +--ro candidate-path
      |        ...
      +--ro adj-rib-in-post
      |  +--ro routes
      |     +--ro route* [distinguisher color endpoint path-id]
      |        ...
      +--ro adj-rib-out-pre
      |  +--ro routes
      |     +--ro route* [distinguisher color endpoint path-id]
      |        ...
      +--ro adj-rib-out-post
         +--ro routes
            +--ro route* [distinguisher color endpoint path-id]
               ...
```

Next Steps

- **Request more review to improve these drafts**.
- **Any comments welcomed.**

Thanks!